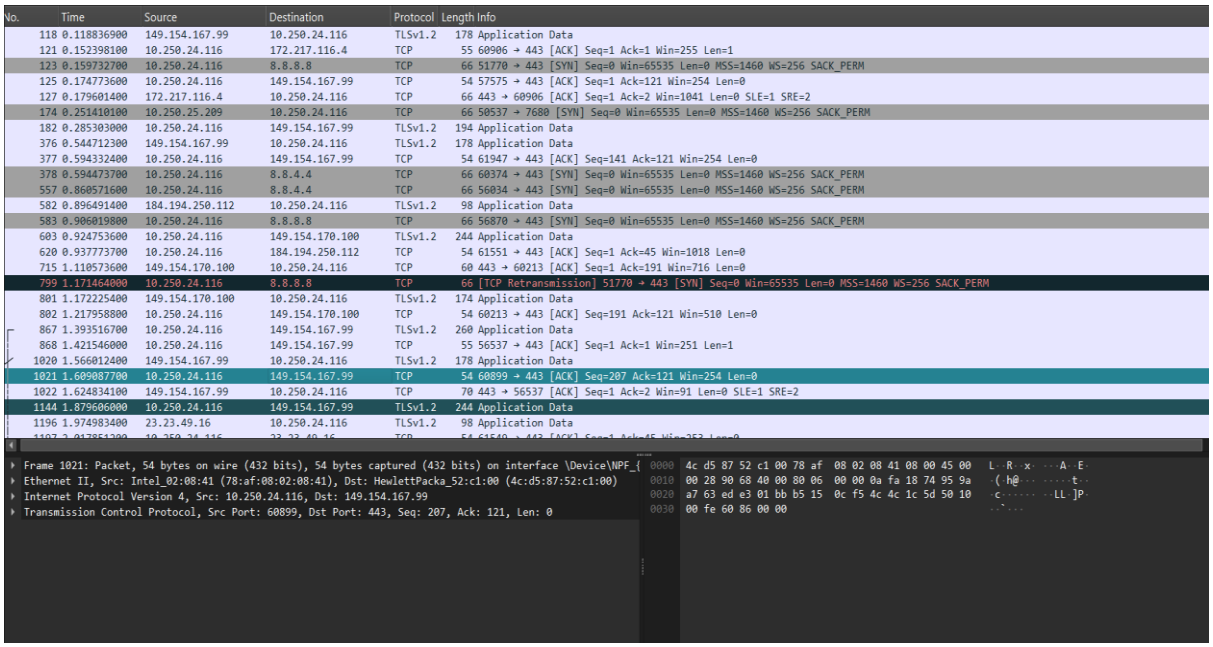
WEEK 1:

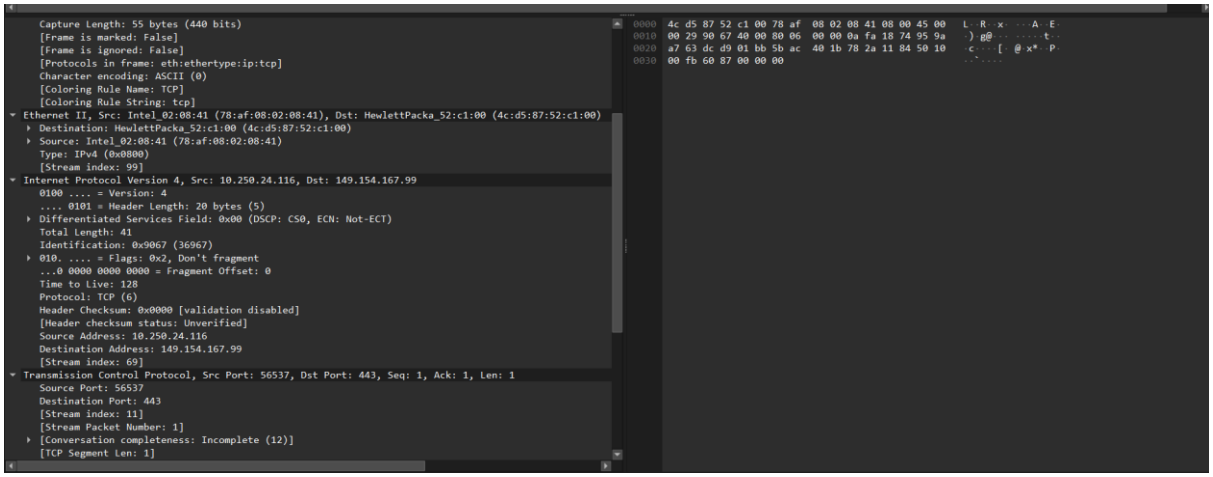
Roll no:2520090194

Name: Naman Masineni

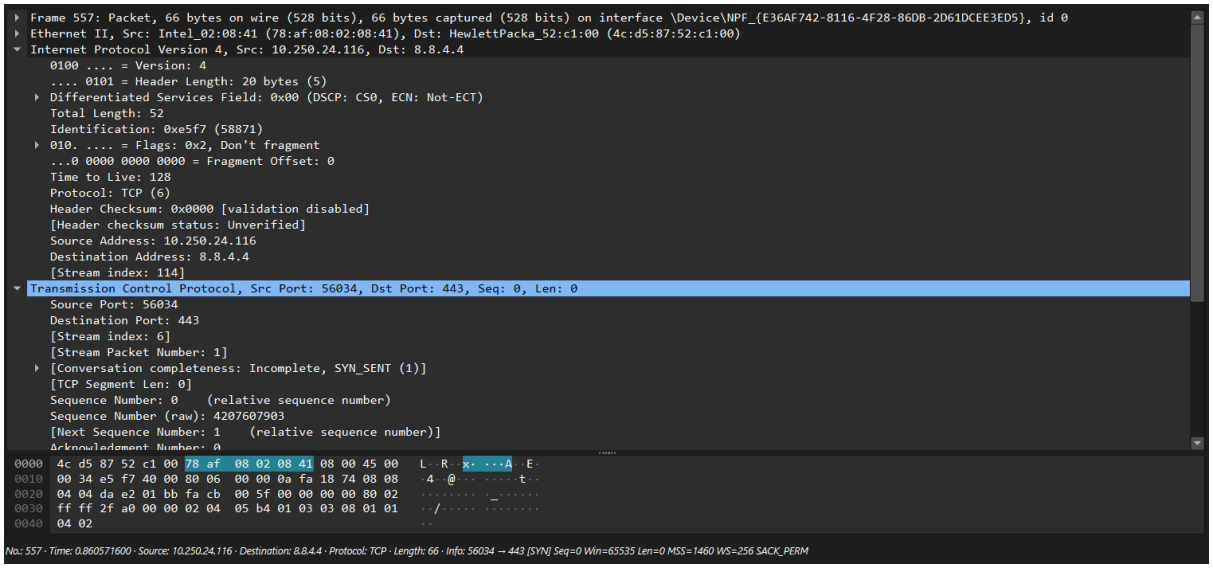
Branch: CSIT

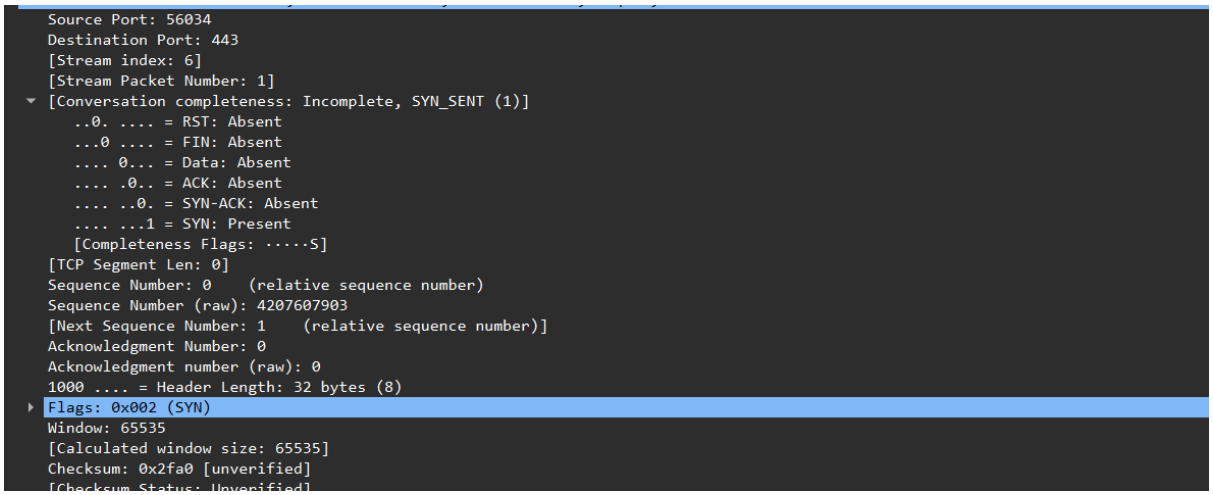
1)TCP FILTER:



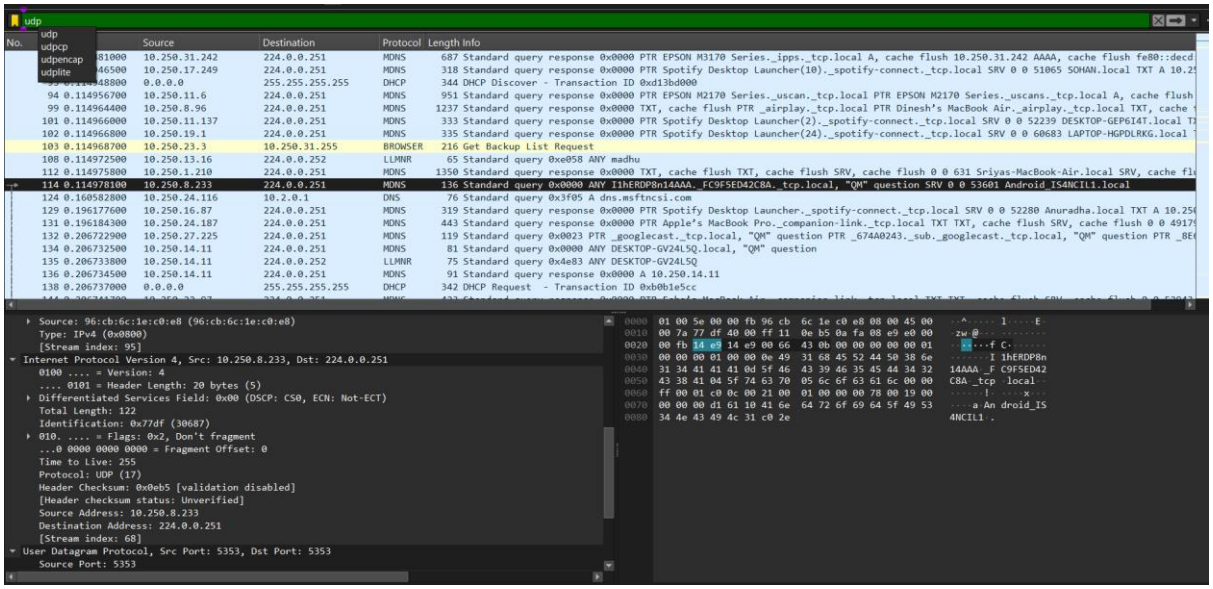


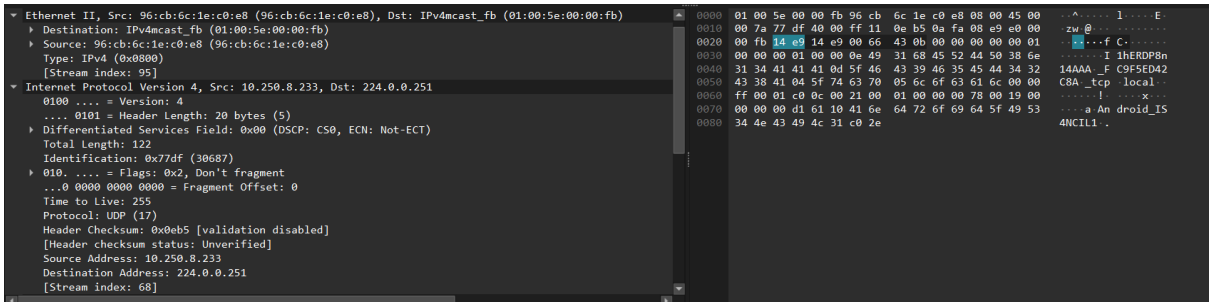
TCP header inspection:

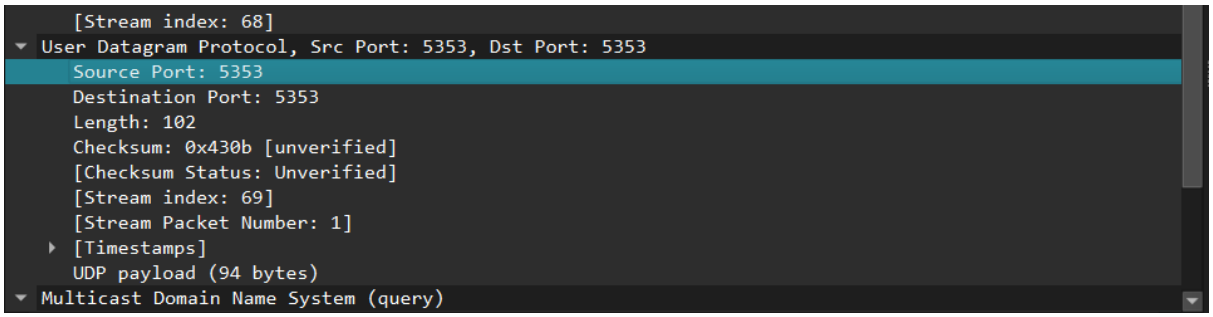

Frame 557: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{E36AF742-8116-4F28-86DB-2D61DCEE3ED5}, id 0
Ethernet II, Src: Intel_02:08:41 (78:af:08:02:08:41), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Internet Protocol Version 4, Src: 10.250.24.116, Dst: 8.8.4.4
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0xe5f7 (58871)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.24.116
Destination Address: 8.8.4.4
[Stream index: 114]
Transmission Control Protocol, Src Port: 56034, Dst Port: 443, Seq: 0, Len: 0
Source Port: 56034
Destination Port: 443
[Stream index: 6]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete, SYN_SENT (1)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 4207607903
[Next Sequence Number: 1 (relative sequence number)]
0000 4c d5 87 52 c1 00 78 af 08 02 08 41 08 00 45 00
0010 00 34 e5 f7 40 00 80 06 00 00 0a fa 18 74 08 08
0020 04 04 da e2 01 bb fa cb 00 5f 00 00 00 00 80 02
0030 ff ff 2f a0 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02
No.: 557 · Time: 0.860571600 · Source: 10.250.24.116 · Destination: 8.8.4.4 · Protocol: TCP · Length: 66 · Info: 56034 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM


Source Port: 56034
Destination Port: 443
[Stream index: 6]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete, SYN_SENT (1)]
..0. .... = RST: Absent
...0 .... = FIN: Absent
.... 0... = Data: Absent
.... .0.. = ACK: Absent
.... ..0. = SYN-ACK: Absent
....1 = SYN: Present
[Completeness Flags: ·····S]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 4207607903
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 .... = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
Window: 65535
[Calculated window size: 65535]
Checksum: 0x2fa0 [unverified]

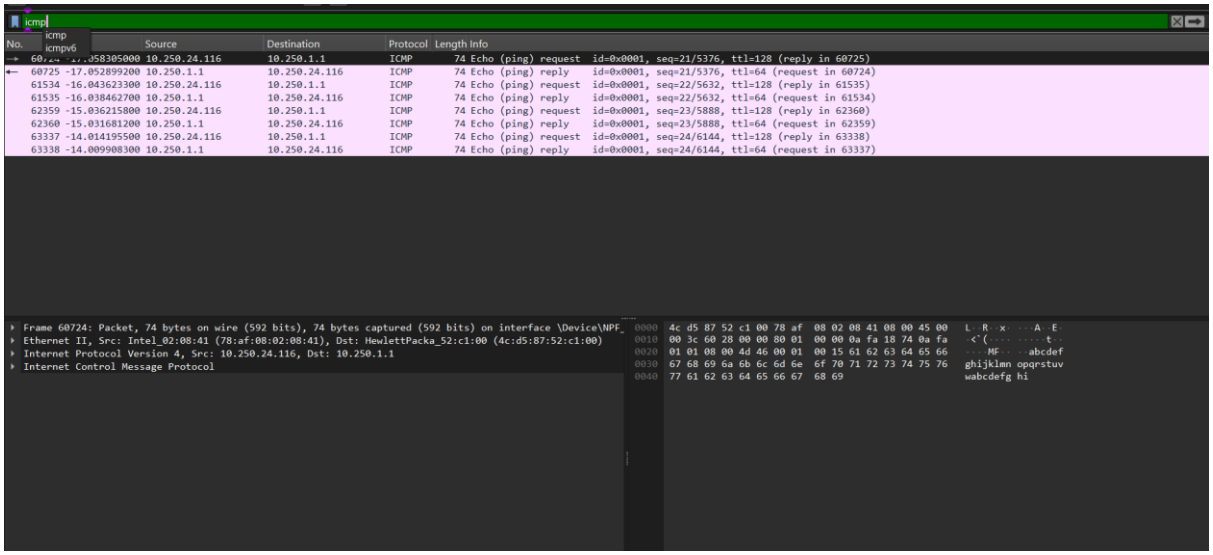
2)UDP :

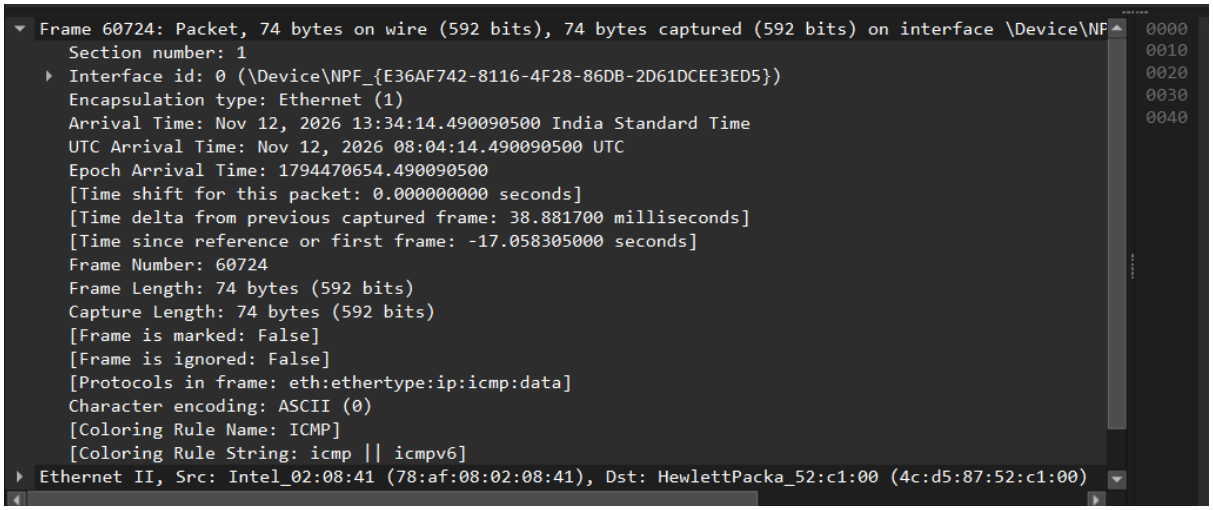

udp
udp
udpcp
udpencap
udplite
No. Source Destination Protocol Length Info
94 0.114956700 10.250.11.6 224.0.0.251 MDNS 951 Standard query response 0x0000 PTR EPSON M2170 Series._uscan._tcp.local PTR EPSON M2170 Series._uscans._tcp.local A, cache flush
99 0.114964400 10.250.8.96 224.0.0.251 MDNS 1237 Standard query response 0x0000 TXT, cache flush PTR _airplay._tcp.local PTR Dinesh's MacBook Air._airplay._tcp.local TXT, cache
101 0.114966000 10.250.11.137 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(2)._spotify-connect._tcp.local SRV 0 0 52239 DESKTOP-GEP6I4T.local
102 0.114966800 10.250.19.1 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Launcher(24)._spotify-connect._tcp.local SRV 0 0 60683 LAPTOP-HGPDLRKG.local
103 0.114968700 10.250.23.3 10.250.31.255 BROWSER 216 Get Backup List Request
108 0.114972500 10.250.13.16 224.0.0.252 LLMNR 65 Standard query 0xe058 ANY madhu
112 0.114975800 10.250.1.210 224.0.0.251 MDNS 1350 Standard query response 0x0000 TXT, cache flush TXT, cache flush SRV, cache flush 0 0 631 Sriyas-MacBook-Air.local SRV, cache fl
114 0.114978100 10.250.8.233 224.0.0.251 MDNS 136 Standard query 0x0000 ANY I1hERDP8n14AAA._FC9F5ED42C8A._tcp.local, "QM" question SRV 0 0 53601 Android_IS4NCIL1.local
124 0.160582800 10.250.24.116 10.2.0.1 DNS 76 Standard query 0x3f05 A dns.msftncsi.com
129 0.196177600 10.250.16.87 224.0.0.251 MDNS 319 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 52280 Anuradha.local TXT A 10.25
131 0.196184300 10.250.24.187 224.0.0.251 MDNS 443 Standard query response 0x0000 PTR Apple's MacBook Pro._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 4917
132 0.206722900 10.250.27.225 224.0.0.251 MDNS 119 Standard query 0x0023 PTR _googlecast._tcp.local, "QM" question PTR _674A0243._sub._googlecast._tcp.local, "QM" question PTR _BE
134 0.206732500 10.250.14.11 224.0.0.251 MDNS 81 Standard query 0x0000 ANY DESKTOP-GV24L5Q.local, "QM" question
135 0.206733800 10.250.14.11 224.0.0.252 LLMNR 75 Standard query 0x4e83 ANY DESKTOP-GV24L5Q
136 0.206734500 10.250.14.11 224.0.0.251 MDNS 91 Standard query response 0x0000 A 10.250.14.11
138 0.206737000 0.0.0.0 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0xb0b1e5cc
Source: 96:cb:6c:1e:c0:e8 (96:cb:6c:1e:c0:e8)
Type: IPv4 (0x0800)
[Stream index: 95]
Internet Protocol Version 4, Src: 10.250.8.233, Dst: 224.0.0.251
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 122
Identification: 0x77df (30687)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 255
Protocol: UDP (17)
Header Checksum: 0x0eb5 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.8.233
Destination Address: 224.0.0.251
[Stream index: 68]
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Source Port: 5353
0000 01 00 5e 00 00 fb 96 cb 6c 1e c0 e8 08 00 45 00
0010 00 7a 77 df 40 00 ff 11 0e b5 0a fa 08 e9 e0 00
0020 00 fb 14 e9 14 e9 00 66 43 0b 00 00 00 00 00 01
0030 00 00 00 01 00 00 0e 49 31 68 45 52 44 50 38 6e
0040 31 34 41 41 41 0d 5f 46 43 39 46 35 45 44 34 32
0050 43 38 41 04 5f 74 63 70 05 6c 6f 63 61 6c 00 00
0060 ff 00 01 c0 0c 00 21 00 01 00 00 00 78 00 19 00
0070 00 00 00 d1 61 10 41 6e 64 72 6f 69 64 5f 49 53
0080 34 4e 43 49 4c 31 c0 2e

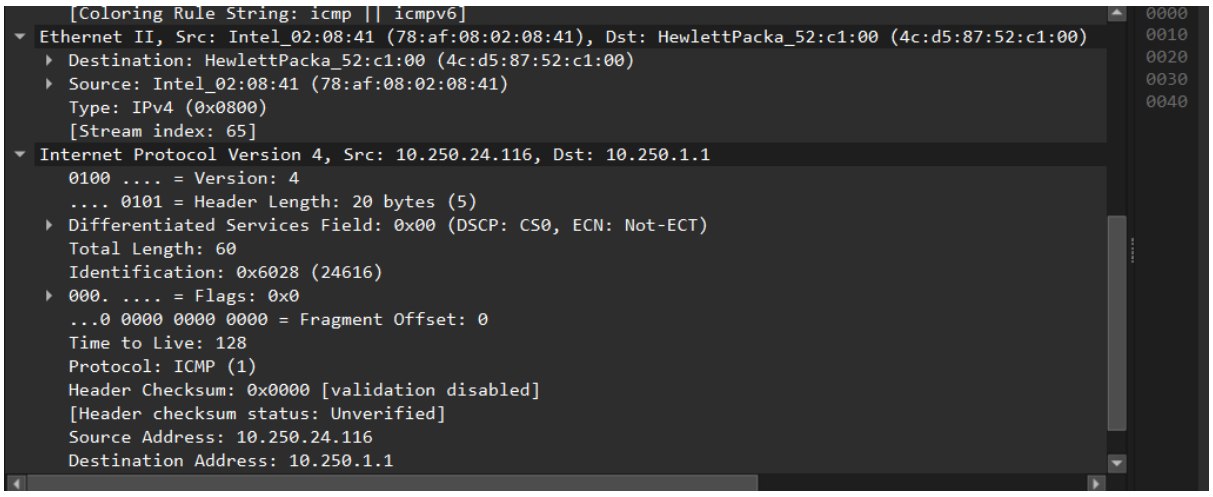

Ethernet II, Src: 96:cb:6c:1e:c0:e8 (96:cb:6c:1e:c0:e8), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
Destination: IPv4mcast_fb (01:00:5e:00:00:fb)
Source: 96:cb:6c:1e:c0:e8 (96:cb:6c:1e:c0:e8)
Type: IPv4 (0x0800)
[Stream index: 95]
Internet Protocol Version 4, Src: 10.250.8.233, Dst: 224.0.0.251
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 122
Identification: 0x77df (30687)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 255
Protocol: UDP (17)
Header Checksum: 0x0eb5 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.8.233
Destination Address: 224.0.0.251
[Stream index: 68]
0000 01 00 5e 00 00 fb 96 cb 6c 1e c0 e8 08 00 45 00
0010 00 7a 77 df 40 00 ff 11 0e b5 0a fa 08 e9 e0 00
0020 00 fb 14 e9 14 e9 00 66 43 0b 00 00 00 00 00 01
0030 00 00 00 01 00 00 0e 49 31 68 45 52 44 50 38 6e
0040 31 34 41 41 41 0d 5f 46 43 39 46 35 45 44 34 32
0050 43 38 41 04 5f 74 63 70 05 6c 6f 63 61 6c 00 00
0060 ff 00 01 c0 0c 00 21 00 01 00 00 00 78 00 19 00
0070 00 00 00 d1 61 10 41 6e 64 72 6f 69 64 5f 49 53
0080 34 4e 43 49 4c 31 c0 2e

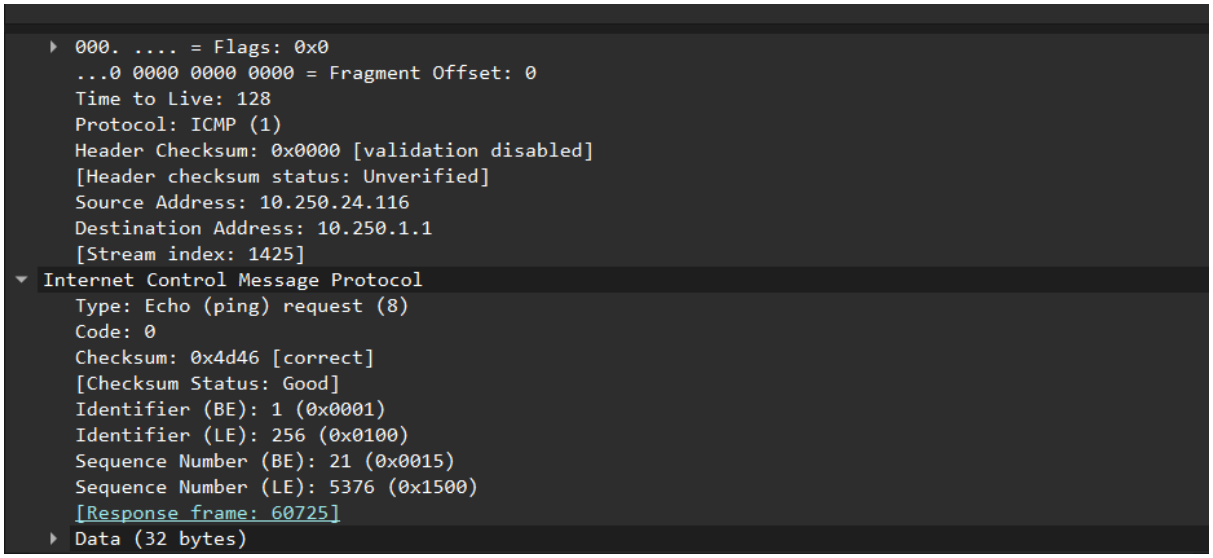

[Stream index: 68]
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Source Port: 5353
Destination Port: 5353
Length: 102
Checksum: 0x430b [unverified]
[Checksum Status: Unverified]
[Stream index: 69]
[Stream Packet Number: 1]
[Timestamps]
UDP payload (94 bytes)
Multicast Domain Name System (query)

ICMP FILTER:


icmp
icmp
icmpv6
No. Source Destination Protocol Length Info
60725 -17.052899200 10.250.1.1 10.250.24.116 ICMP 74 Echo (ping) reply id=0x0001, seq=21/5376, ttl=64 (request in 60724)
61534 -16.043623300 10.250.24.116 10.250.1.1 ICMP 74 Echo (ping) request id=0x0001, seq=22/5632, ttl=128 (reply in 61535)
61535 -16.038462700 10.250.1.1 10.250.24.116 ICMP 74 Echo (ping) reply id=0x0001, seq=22/5632, ttl=64 (request in 61534)
62359 -15.036215800 10.250.24.116 10.250.1.1 ICMP 74 Echo (ping) request id=0x0001, seq=23/5888, ttl=128 (reply in 62360)
62360 -15.031681200 10.250.1.1 10.250.24.116 ICMP 74 Echo (ping) reply id=0x0001, seq=23/5888, ttl=64 (request in 62359)
63337 -14.014195500 10.250.24.116 10.250.1.1 ICMP 74 Echo (ping) request id=0x0001, seq=24/6144, ttl=128 (reply in 63338)
63338 -14.009908300 10.250.1.1 10.250.24.116 ICMP 74 Echo (ping) reply id=0x0001, seq=24/6144, ttl=64 (request in 63337)
Frame 60724: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_
Ethernet II, Src: Intel_02:08:41 (78:af:08:02:08:41), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Internet Protocol Version 4, Src: 10.250.24.116, Dst: 10.250.1.1
Internet Control Message Protocol
0000 4c d5 87 52 c1 00 78 af 08 02 08 41 08 00 45 00
0010 00 3c 60 28 00 00 80 01 00 00 0a fa 18 74 0a fa
0020 01 01 08 00 4d 46 00 01 00 15 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69
ghijklmn opqrstuv
wabcdefg hi


Frame 60724: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NP
Section number: 1
Interface id: 0 (\Device\NPF_{E36AF742-8116-4F28-86DB-2D61DCEE3ED5})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 12, 2026 13:34:14.490090500 India Standard Time
UTC Arrival Time: Nov 12, 2026 08:04:14.490090500 UTC
Epoch Arrival Time: 1794470654.490090500
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 38.881700 milliseconds]
[Time since reference or first frame: -17.058305000 seconds]
Frame Number: 60724
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
Character encoding: ASCII (0)
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
Ethernet II, Src: Intel_02:08:41 (78:af:08:02:08:41), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
0000
0010
0020
0030
0040


[Coloring Rule String: icmp || icmpv6]
Ethernet II, Src: Intel_02:08:41 (78:af:08:02:08:41), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Source: Intel_02:08:41 (78:af:08:02:08:41)
Type: IPv4 (0x0800)
[Stream index: 65]
Internet Protocol Version 4, Src: 10.250.24.116, Dst: 10.250.1.1
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 60
Identification: 0x6028 (24616)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.24.116
Destination Address: 10.250.1.1
0000
0010
0020
0030
0040



Parameter	TCP Packet	UDP Packet
Frame Length	533 bytes	106 bytes
Ethernet Header Length	14 bytes	14 bytes (+4B VLAN)
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	519 bytes	88 bytes
Transport Header Length	20 bytes	8 bytes
Payload Length	479 bytes	60 bytes

1.What is encapsulation? Encapsulation is the process where each protocol layer wraps application data with its own control information (header, and sometimes a trailer) as the data moves down the network stack before transmission

2. What is the purpose of the Ethernet header? The Ethernet header (Layer 2) enables local network delivery by specifying the source and destination MAC addresses and identifying the encapsulated upper-layer protocol (such as IPv4 or ARP) via the EtherType field.

3. What information is stored in the IP header? The IP header (Layer 3) stores logical addressing and routing details, including the source and destination IP addresses, IP version, header length, total packet length, Time to Live (TTL), protocol number (e.g., TCP or UDP), fragmentation flags, and a header checksum.

4. What is the difference between TCP and UDP headers?

- TCP header: Typically 20–60 bytes long; includes sequence numbers, acknowledgment numbers, window size, and control flags (SYN, ACK, FIN) to guarantee reliable, ordered, connection-oriented delivery.

- UDP header: Fixed at 8 bytes; contains only source port, destination port, length, and checksum, offering fast, connectionless, best-effort delivery with minimal overhead.

5.Why does each protocol add a header? Each protocol layer adds a header to provide the specific addressing, routing, error checking, sequencing, and delivery instructions needed for its layer to function independently and hand off data properly at the receiving device.

6. What is packet overhead? Packet overhead is the total number of extra control bytes (headers and trailers) added across protocol layers that do not carry actual user or application payload data.

7. How does header overhead affect network efficiency? Header overhead consumes bandwidth and transmission capacity without carrying user data, lowering payload efficiency ($\frac{\text{Payload}}{\text{Total Frame Size}} \times 100$). Efficiency drops significantly when sending many small payloads because the fixed header bytes make up a large percentage of each frame.